

Anusha Ravindra

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PROFESSIONAL SUMMARY

Electrical engineer with expertise in electric machine design, FEA modeling, and simulation automation. Developed rare-earth-free motor technologies achieving significant material cost savings and reduced design cycle time substantially.

EDUCATION

- Texas A&M University, College Station, TX** United States
Master of Science in Electrical Engineering; GPA: 4.0 May 2027
Coursework: Power Electronics, DSP-Based Electromechanical Motion Control, Electronic Motor Drives, Linear Multivariable Systems, Computer Aided Design of Electromechanical Motion Devices
- Sapthagiri College Of Engineering, Karnataka** India
Bachelor of Technology in Electrical and Electronics Engineering; GPA: 3.628 June 2023

SKILLS

- Programming Languages:** Python, MATLAB, PyAnsys, C++.
- Software:** AutoCAD, SIMULINK, Ansys Maxwell 2D, Ansys Maxwell 3D, Ansys MotorCad, Solidworks, Simcenter MotorSolve, Simcenter Magnet, PLECS, Altium, ElmerFEM, Gmsh, MultiSim, Microsoft Excel, PowerPoint, Word.
- Analytical Concepts & Coursework:** Electromagnetic Simulation and Optimization, Finite Element Analysis (FEA), Electric Motor Design, Automation of Simulation Workflows, Technical Documentation, Performance Verification against Simulated Data, 3D Printing.

EXPERIENCE

- Toshiba International Corporation** Houston, TX, United States
R&D Electrical Engineering Intern May 2026 – Present
 - Developed a macro-free IEEE 112 Method F1 induction motor test reporting tool using Excel LAMBDA functions to automate iterative equivalent-circuit calculations and report generation.
 - Redesigned Method F1 input and output sheets to simplify test-data entry, improve engineering review, and streamline customer report preparation.
 - Developed a configurable nameplate compliance validation tool for 5,000+ industrial motor rating IDs, identifying data discrepancies and verifying motor specifications against master databases, NEMA Premium, and DOE requirements.
 - Built an automated lead-wire length lookup database and part-number-based Excel interface, enabling manufacturing teams to quickly retrieve lead-wire lengths across multiple motor configurations.
- Advanced Electrical Machine and Power Electronics Lab, Texas A&M University** Texas, United States
Student Graduate Worker September 2025 – Present
 - Supporting research on the design and electromagnetic modeling of an Asynchronously Excited Brushless Wound-Field Synchronous Machine.
 - Performing FEA-based electromagnetic analysis, performance evaluation, and design optimization.
 - Developing code to interpolate datasets and extract parameters from lookup tables for motor analysis and control.
 - Working on the logic and code to analyse data and plot Torque-speed Curve and Efficiency Map of an Asynchronously Excited Brushless Wound-Field Synchronous Machine.
- Chara Technologies Pvt. Ltd** Karnataka, India
Electromagnetic Design Engineer June 2023 – June 2025
 - Designed rare-earth-free motors achieving 40% cost reduction while maintaining efficiency.
 - Automated motor simulation workflows using Python and MATLAB for Ansys Maxwell and Simcenter Magnet.
 - Implemented open-source simulation pipelines using Gmsh and ElmerFEM, reducing licensing costs by 90%.
 - Developed multi-objective optimization algorithms including Gradient Descent and Genetic Algorithms.
 - Designed, prototyped, and tested a 5HP ferrite-assisted SynRM achieving IE5 efficiency.
 - Applied field-oriented control (FOC) concepts for synchronous reluctance motors.

PATENT

Filed a patent application for “**Novel motor rotor design with peripherally disconnected flux carriers of Ferrite Assisted Synchronous Reluctance Motors to enhance efficiency and minimize rotor iron loss**”. Currently under review.